

Proposed Solution

Date: 11 June 2026

Project Name: E-commerce Application

Proposed Solution Details

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Customers need a secure, fast, and intuitive platform to browse and purchase items online, while administrators need an efficient tool to manage inventory, sales, and order tracking.
2	Idea / Solution description	Develop a responsive Single Page Application (SPA) using the MERN stack (MongoDB, Express, React, Node.js). The platform will feature robust authentication, full-text search, shopping cart management, seamless checkout via Payment Gateway (Planned), and an administrative dashboard for business operations.
3	Novelty / Uniqueness	The application couples decoupled architecture with modern UX practices. Unlike monolithic alternatives, the React frontend allows for micro-animations and instantaneous cart updates, while the Node.js backend ensures scalable API responses. Integration of Cloudinary allows dynamic image transformations without overloading the local server.
4	Social Impact / Customer Satisfaction	Provides a reliable and transparent purchasing process that builds consumer trust. Small business owners are empowered with accessible analytics, allowing them to scale their operations efficiently and improve customer service based on review feedback.
5	Business Model (Revenue Model)	Primary revenue will be generated through direct sales margins on products. Secondary models could include premium vendor placements (if multi-vendor), or shipping and handling fees.